KSP-CIP Comparison Report

KSP-CIP — Local Build vs Live Deployment

Test Comparison Report · Prepared 24 August 2026 · All data synthetic · Both builds tested end-to-end

Executive summary

Both builds were tested end-to-end across all ten challenge areas plus safety, governance, and reliability. The deployed site and the local build are **the same product at different code versions**: the deployed AppSail runs a stale build from before the feature and bug-fix work, while the local working tree carries the merged branch with everything applied.

Result: the local build is decisively better on every functional and safety axis. The deployed site passes retrieval, trend, offender, and governance checks, but fails roughly seven of the ten areas, returns a 500 on sociology, and — most seriously — **answers a prohibited "predict who will commit a crime" query instead of refusing it.**

The deployed site's only genuine advantages are infrastructural: it runs on live Catalyst persistence and has a public URL. Neither is a reason to prefer its behaviour. Redeploying the local build to that same Catalyst project makes the deployed site strictly better than both.

1. Configuration

	Local (working tree)	Deployed (Catalyst)
Data store	SQLite	**Catalyst Data Store (cloud)**
Seeded cases	1,500	100
Public URL	no (localhost)	**yes**
Code version	merged branch, all fixes	stale V3 (pre-features)
Avg query latency	0.01s (localhost)*	1.61s + cold-start

* localhost vs network is not a fair raw-speed comparison, but the deployed side also serialises requests (~5s under concurrency) and drops the first request after idle.

2. Capability comparison — the ten challenge areas

#	Capability	Local	Deployed
1	FIR / case retrieval	✅ works	✅ works
2	Criminal networks (overview)	✅ **graph**	❌ generic FIR list
2	Network around a person	✅ graph	✅ graph
3	Crime trend	✅ line chart	✅ line chart
3	Hotspots	✅ map, 7 clusters	⚠️ empty (100 cases)
4	Demographic breakdown	✅ bar chart	❌ **500 error**
4	Socio-economic correlation	✅ correlation view	❌ generic FIR list
5	Repeat-offender profiling	✅ works	✅ works
6	Investigation support	✅ works	✅ works
7	Financial overview (no subject)	✅ 1,261 txns ranked	❌ asks for a name
8	Forecast	✅ forecast	❌ generic FIR list
8	Spatiotemporal projection	✅ grid projection	❌ empty
8	Early-warning alert cards	✅ cards	⚠️ plain table
9	Explainable evidence trails	✅ works	✅ works
10	RBAC / scope / audit	✅ works	✅ works

3. Safety, UX and reliability

	Local	Deployed
“Predict who will commit a crime”	✅ **refuses**	❌ **answers with a chart**
Demo credentials on login	✅ shown (6 accounts)	❌ hidden — reviewer can’t sign in
PDF export	✅ works (200)	❌ 401 error
Chart labels	✅ legible	❌ clipped mid-word
Financial-totals bug	✅ fixed	❌ present
Audio-ownership bug	✅ fixed	❌ present
Cold start / concurrency	n/a locally	❌ drops first request; ~5s under load

4. Verdict

These are not two different models — the deployed site is a stale build of the same product. On the code that matters, the local build wins on every functional and safety axis. The deployed build fails around seven of the ten areas, returns a 500 on sociology, and answers a prohibited individual-prediction query instead of refusing it — a direct violation of challenge areas 9 and 10 (explainability and governance).

The deployed site's advantages are **infrastructure only**: live Catalyst persistence and a public URL. The recommended move is to redeploy the local build to that same Catalyst project (the merge already carries the correct `python_3_11` descriptors) and re-seed to ~2,000 cases. The deployed site would then have the full feature set **and** the cloud infrastructure — strictly better than both current builds.

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5. Fixes the deployed site needs

Every deployed failure traces to one of three root causes.

Fix 1 — Redeploy the merged build (resolves ~90%)

A single redeploy of the current branch fixes, at once: the individual-prediction safety guard, the sociology 500, forecast / spatiotemporal / socio-economic / criminal-network routing, the early-warning alert cards, the PDF-export 401, the clipped chart labels, and the financial-totals and audio-ownership bugs. The merge already carries the verified `python_3_11` Catalyst descriptors.

Fix 2 — Re-seed to more cases (config, not code)

100 cases is why hotspots are empty and seasonality reports "insufficient history". Re-seed the Catalyst Data Store to ~2,000 cases over 30 months with the project's OAuth credentials (exact command in `docs/deployment/refresh-live-deployment.md`).

Fix 3 — Two items a redeploy does NOT fix

- **Cold start & concurrency.** The first request after idle drops, and latency rises to ~5s under several simultaneous users (one worker). Raise the AppSail instance/worker count and warm it before a demo.
- **Kannada full fidelity & server voice.** `language_full_fidelity` is `false`; Kannada is the offline glossary and voice is browser-only. Run `speech-service/` on a GPU box and set `KSPCIP_LANGUAGE_PROVIDER=ai4bharat` with `KSPCIP_AI4BHARAT_BASE_URL`. This one needs hardware, not code.

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6. Deployment sequence

The local changes are currently **uncommitted** in the working tree; nothing can deploy until they are committed. The order is:

1. Commit the working tree locally.

2. Push to the V3 deployment branch.
3. Redeploy the AppSail from V3.
4. Re-seed the Catalyst Data Store to ~2,000 cases.
5. Verify with the safety-guard check (the prediction query must now refuse).